

Aditya Chunduri

Los Angeles, CA | (678) 776-0805 | chunduri@usc.edu | LinkedIn | GitHub | Portfolio

Summary

New graduate seeking a Machine Learning / GenAI Engineer role focused on RAG systems and LLM evaluation. Built end-to-end LLM pipelines spanning retrieval, prompting, and quality/guardrail workflows, with an emphasis on reproducible experiments (JSON-driven tests, dashboards) and privacy-aware architectures. Hands-on with PyTorch, LangChain, Hugging Face, and vector databases (Pinecone/Chroma), motivated to ship trustworthy AI features from prototype to production.

Technical Skills

- **Languages & Frameworks:** Python, SQL, C++, PyTorch, TensorFlow, Scikit-learn, LangChain, Hugging Face
- **GenAI & ML Systems:** LLM Evaluation, RAG Architectures, Prompt Engineering, Multimodal ML, Model Benchmarking, Human-in-the-Loop Systems
- **Data, Audio & Vision:** Pandas, NumPy, Spark, Librosa, Whisper, Demucs, OpenCV, CNNs, U-Net, YOLOv5
- **Databases & Infra:** PostgreSQL, Pinecone, ChromaDB, SQLAlchemy, AWS SageMaker, Azure
- **Tools & Platforms:** Git, GitHub, Docker, Flask, Gradio, Selenium, UiPath, PowerPlatform, Power BI Automate, Power BI, Virtual Agents)

System-Level Projects

LLM Evaluation & Analysis Platform – Model Behavior Lab Jul 2025 – Aug 2025

- Designed and built an automated evaluation framework to benchmark LLMs across reasoning accuracy, hallucination propensity, emotional alignment consistency, and code correctness.
- Implemented JSON-driven test pipelines enabling reproducible experiments across Phi-3, Mistral, and Samantha models.
- Developed visualization dashboards to surface behavioral tradeoffs, failure modes, and consistency gaps across models.
- Used evaluation results to inform prompt strategies and deployment-oriented model selection for downstream GenAI applications.

Retrieval-Augmented LLM Insight Engine – AI Health Journal May 2025 – Jun 2025

- Built a privacy-aware journaling assistant using LangChain, Pinecone, and Flask to generate structured mental-health insights.
- Implemented RAG pipelines to ground responses in the user-specific historical context, reducing the risk of hallucination in personalized responses.
- Integrated Whisper-based voice transcription, enabling a seamless voice-to-analysis workflow.
- Designed prompts and retrieval logic to balance empathy, factual grounding, and interpretability.

Multimodal Audio Intelligence System – AI RemixMate Jan 2025 – Apr 2025

- Engineered an intelligent music remixing system combining audio signal processing with ML-driven tempo and harmonic alignment.
- Extracted MFCCs, tempo, and harmonic features to align tracks algorithmically before remixing.
- Integrated Demucs for stem separation and Whisper for lyrical alignment.
- Achieved ~90% perceived transition smoothness across 50+ evaluated remixes through iterative testing.

Automated GenAI Media Pipeline – Akashic Tree Aug 2025 – Present

- Built a modular brief→script→image→voice→video pipeline with intermediate artifacts saved per stage.
- Implemented brand/style constraints and validation checks to catch off-brand scripts before downstream generation.
- Added quality gates with early-exit behavior to block low-quality generations and reduce wasted compute resources.
- Integrated Stable Diffusion (images), Coqui TTS (voice), and MoviePy (assembly) with config-driven batch runs and logging.

Natural Language to SQL / NoSQL Query Engine – ChatDB Aug 2024 – Dec 2024

- Built an NLP-driven query system translating natural language into optimized database queries.
- Combined spaCy-based parsing with schema-aware retrieval to improve intent resolution and query correctness.
- Prototyped a RAG-based assistant to assist developers with query formulation.
- Improved translation accuracy by ~25% through iterative refinement.

Research Experience

University of Southern California Los Angeles, CA

Research Assistant – Computer Vision Aug 2024 – Dec 2024

- Built data preprocessing and segmentation pipelines in Python using TensorFlow for over 10,000 retinal fundus images, reducing manual preparation time by 30% and enabling faster model training
- Fine-tuned U-Net architectures with TensorFlow and research-backed augmentation strategies, achieving a 15% increase in validation accuracy for disease classification
- Improved downstream disease detection performance by ~20% by integrating a human-in-the-loop validation step and optimizing hyperparameters in Scikit-learn
- Led weekly technical discussions with clinicians and engineers, aligning model metrics to research goals, which streamlined decision-making and accelerated study approvals

SSN College of Engineering Chennai, India

Research Assistant – ML Engineering Jun 2021 – Jul 2021

- Developed YOLOv5-based object detection models for aerial imagery, achieving 92%+ precision on small-object detection tasks.
- Optimized large-scale video preprocessing and ETL pipelines, reducing runtime by ~25%.
- Built interactive visualization dashboards with Plotly Dash and GitHub CI/CD, enabling faster analysis and supporting quicker decision-making for the research team

Education

University of Southern California Los Angeles, CA

M.S. in Applied Data Science Jan 2024 – Dec 2025

- **Coursework:** Machine Learning, Information Retrieval, Data Mining, Analysis of Algorithms, Data Management, Research Methods

SRM Institute of Science and Technology Chennai, India

B.Tech in Computer Science & Engineering June 2019 – July 2023

- **Specialization:** Artificial Intelligence & Machine Learning